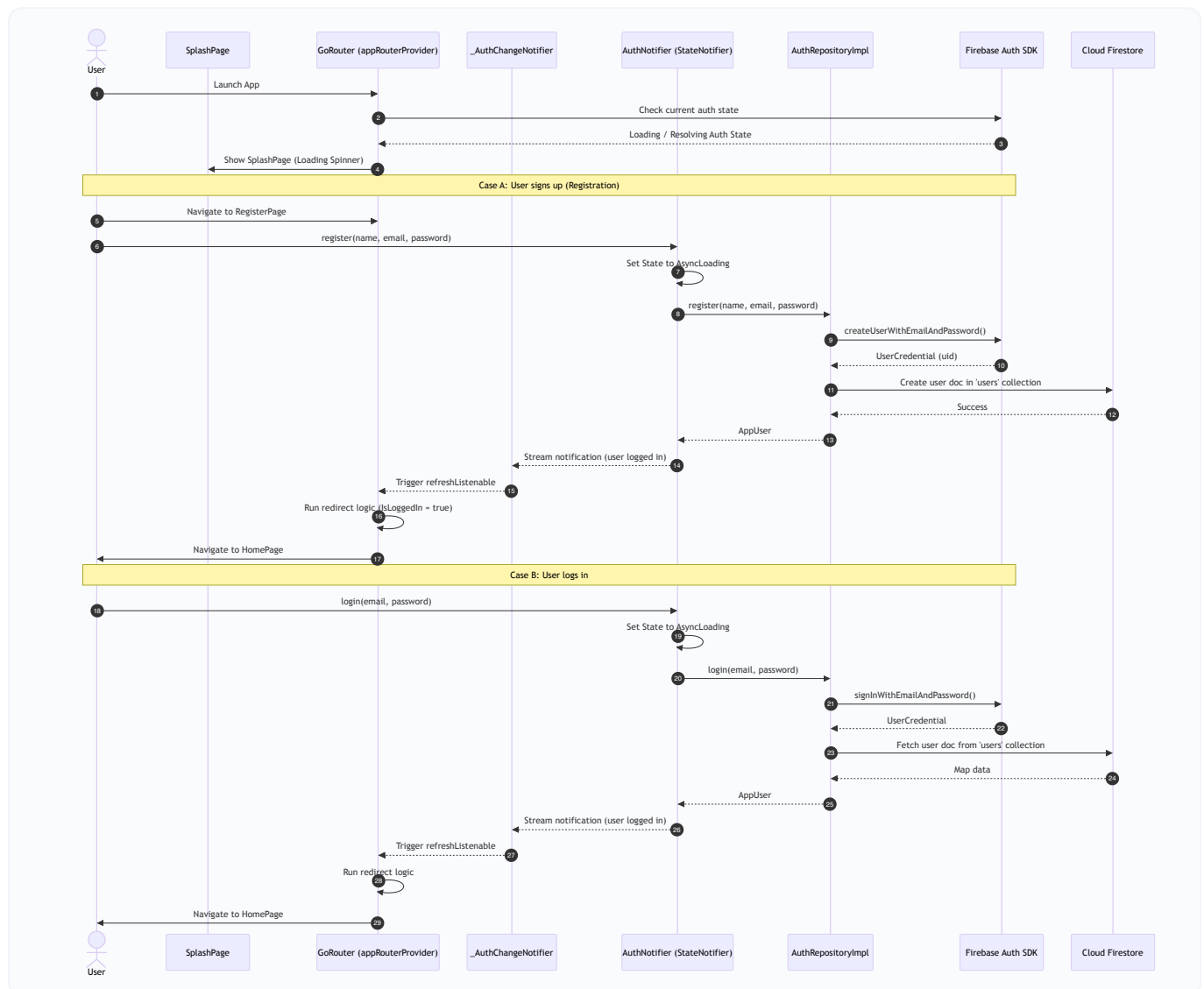


MeetFlow AI — Graduation Project Documentation

SYSTEM SEQUENCE DIAGRAMS

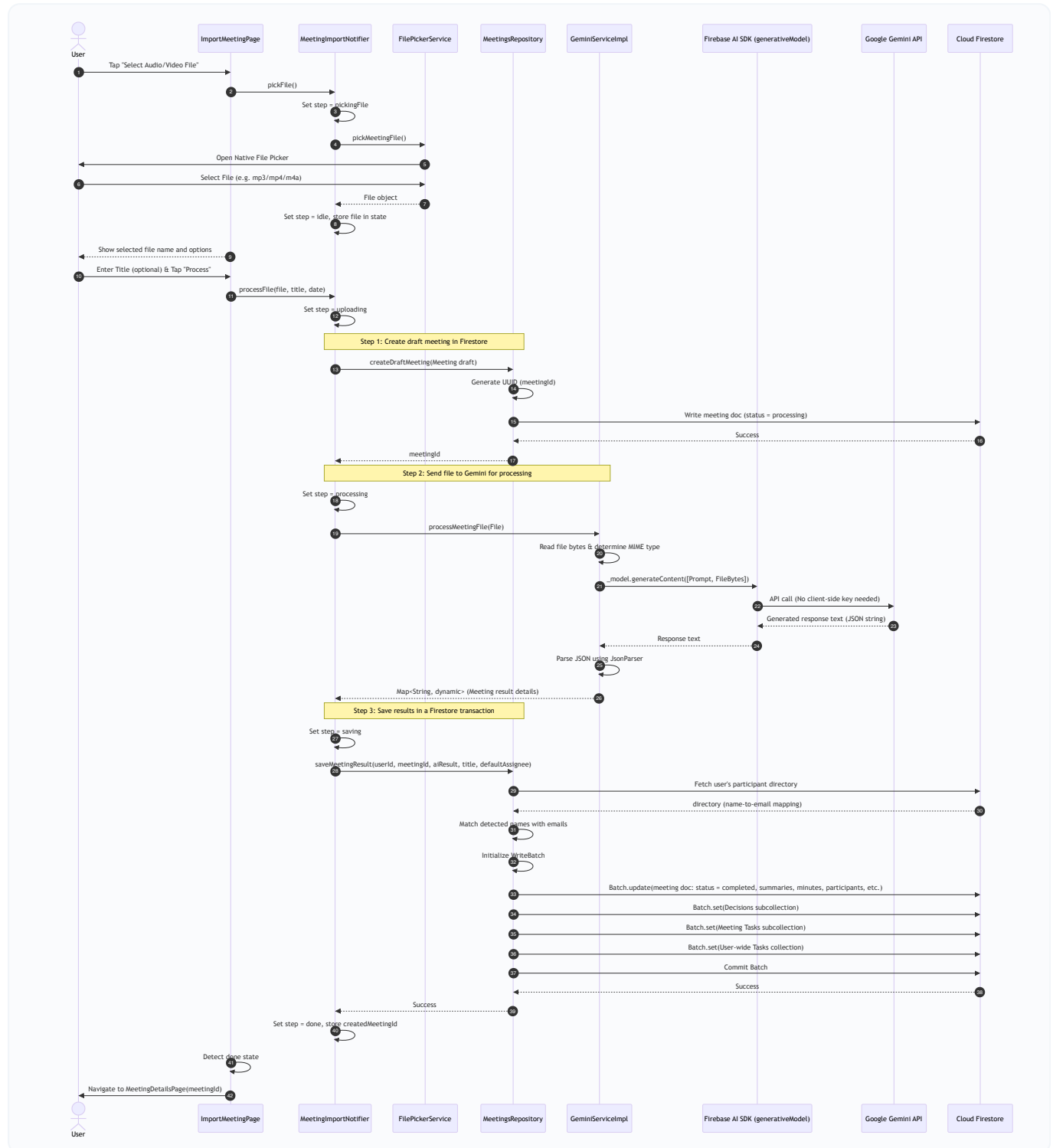
1. User Authentication & Session Routing Flow

This diagram illustrates how user authentication actions (registering and logging in) flow from the presentation layer (using Riverpod StateNotifier) through the repositories to Firebase Authentication and Cloud Firestore, and how GoRouter's refresh listener automatically guards and redirects routes.



2. Audio/Video File Import & AI Meeting Analysis Flow

This diagram shows the end-to-end flow of importing a meeting from a local audio/video file. It describes how a draft meeting is created, uploaded, analyzed by the Google Gemini model using the Firebase AI SDK, parsed from JSON, and saved into Firestore using batch operations to preserve transactional integrity.



3. Task Management, Assignee Reassignment & Email Distribution Flow

This diagram covers the flow of updating task assignments and launching email dispatch. Composing emails builds a mailto URL using a designated translation (Arabic/English) and invokes url_launcher to redirect to the system email composer.

